HP_power_board

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LP_logic_board

File: LP_logic_board.kicad_sch

pi_power_stub

File: pi_power_stub.kicad_sch

pack_sense_board

File: pack_sense_board.kicad_sch

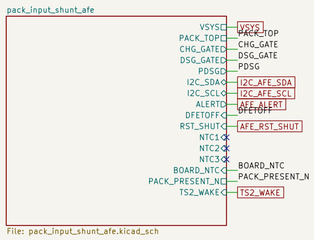
marengo

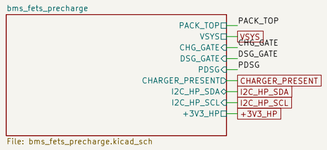
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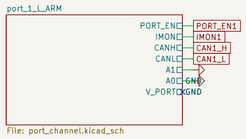
Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 1/27

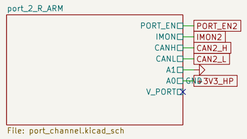


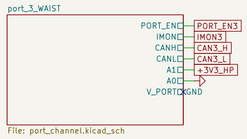


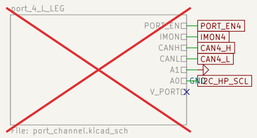


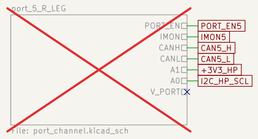


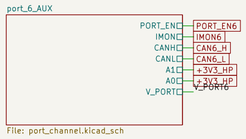




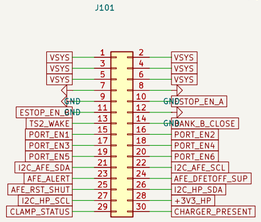




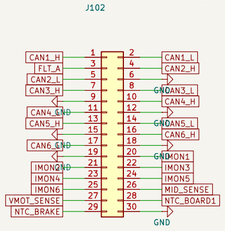




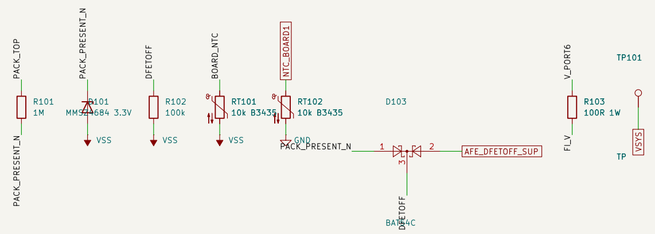
Board-to-board: two 2x15 2 mm ribbons to LP, pin for pin. No stock 2x30 latched header exists. All 60 pins used.

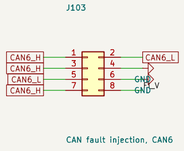


2x15 2 mm box, power and logic



2x15 2 mm box, CAN and analog





CAN fault injection, CAN6

DFETOFF, active high, BOTHOFF (config 0x42): pack sense harness or cell plug pulled, or supervisor backup, opens DSG and CHG. E-stop loop deliberately not wired here: it would deadlock boot (12 V comes after DSG) and drop compute.
Pull-up from PACK_TOP, not LP: it holds with the LP rails down. 54 uA through the presence loop when mated.
Fault injection: test mode, disarmed. Short H-L, H-GND, L-GND, H-VBUS via 100R.

marengo
Sheet: /HP_power_board/
File: HP_power_board.kicad_sch
Title: HP_power_board: high-power board
Size: A2 | Date: 2026-11-06 | **Rev: A-wip**
KiCad E.D.A. 10.0.8 | Id: 2/27

BAT from the pack + terminal, so the AFE stays up if J2 is pulled
with pack power still connected. Fig 16-2: diode, 100R, 1uF, 100pF. CP1 is 1uF, not 0.47uF: see sheet bms_fets_precharge. Caps 100 V.

PACK_TOP D1 BAV19W 120V R15 100R BAT
C14 1uF 100V VSS
C15 100pF VSS
C16 1uF CP1

J2 2x10 locking header
NTC1 NTC3 B0 B2 B4 B6 B8 B10 B12 NTC2 PACK_PRESENT_N B1 B3 B5 B7 B9 B11 B13

J2: harness to the pack sense board 2x10, same pin map.
Columns 4-10 zigzag B0..B13: a bent pin shorts 1 cell
across a column, 2 cells along a row.
Pin 3: NTC and presence return, to VSS here, no cell connection.
Pin 4: B0 Kelvin tap, no bond to BAT- (R below). Spare: pin 12.

J1 pack +, J3 pack -: 10 AWG ring-lug pigtails to the XT90 at the pack.

J1 PACK_TOP M4 terminal, REDCUBE 74651194R
J3 BAT- NT1 M4 terminal, REDCUBE 74651194R
Net-Tie VSS
SRP SRN RS1 0.3mR CSS4J-4026 5W GND

Pack negative -> shunt -> system GND. SRP is the cell side;
charge current makes SRP positive (datasheet p5).
AFE VSS ties at BAT-, so VSS sits above GND by I x 0.3 mR
in discharge: 60 mV at the 200 A SCD point.

B13 R1 20R VC16 C1 220nF
B12 R2 100R VC12 C2 220nF
B11 R3 100R VC11 C3 220nF
B10 R4 100R VC10 C4 220nF
B9 R5 100R VC9 C5 220nF
B8 R6 100R VC8 C6 220nF
B7 R7 100R VC7 C7 220nF
B6 R8 100R VC6 C8 220nF
B5 R9 100R VC5 C9 220nF
B4 R10 100R VC4 C10 220nF
B3 R11 100R VC3 C11 220nF
B2 R12 100R VC2 C12 220nF
B1 R13 100R VC1 C13 220nF
B0 R14 100R VC0

Cell filter: 100R + 220nF, 20R on VC16 (SLUAA81 2.5, top cell IR drop)
(datasheet p9: RC 20-100R, CC 0.1/0.22/1 uF). 100R gives the balance FET its gate drive.

B0 R24 10k BAT- D15 SS14-E3/61T 40V 1A VC0 VSS
C23 1uF 25V X7R B0 VSS

Keeps VC0 defined with the harness out. 9 uA at 60 A.
C23 1 uF to VSS (TI range 0.1 to 1 uF, 100R at its maximum) and D15
hold VC0 through a short: 525 A peak with the 2 uH pack.
B0 to VSS return 1.0 mOhm: -0.22 V, no diode needed.
1.5 mOhm: -0.24 V at 25 C, -0.32 V at -40 C (limit -0.3 V).
Keep the return at 1 mOhm or less.

U1 BQ76952PFBR

BAT	47	45	CHG_GATE
CP1	46	43	DSG_GATE
		40	PCHG
		39	PDSG
		42	PACK
		41	LD
		38	FUSE
		37	BREG
		36	REGIN
		35	REG1
		34	REG2
		24	REG18
VC16	48	27	SDA
VC15	1	26	SCL
VC14	2	25	ALERT
VC13	3	33	RST_SHUT
VC12	4	30	DFETOFF
VC11	5	29	CFETOFF
VC10	6	28	HDQ
VC9	7	31	DCHG
VC8	8	32	DDSG
VC7	9	21	TS1
VC6	10	22	TS2
VC5	11	23	TS3
VC4	12		
VC3	13		
VC2	14		
VC1	15		
VC0	16		
SRP_F	18		
SRN_F	20		
VSS	17		
NC	19		
NC	44		

CHG_GATE DSG_GATE PDSG PACK_S LD_S VSS C19 2.2uF VSS I2C_SDA I2C_SCL ALERT RST_SHUT DFETOFF NTC3 BOARD_NTC NTC1 TS2_F NTC2

13S per Table 16-3 (p75): cells 1, 2, 16 must be real.
VC0..VC12 = cells 1-12. VC15-VC16 = cell 13.
VC13-15 short to VC12. Cell inputs in use: 1-12 and 16.

SRP R16 100R SRP_F
C17 100nF
C18 100pF
SRN R17 100R SRN_F

p65, p76: 100R each side, 100nF + 100pF differential, near U1,
symmetric, same PCB side. Kelvin SRP/SRN routed in parallel from RS1.

PACK and LD both sense VSYS, the common port after the FETs
(Fig 16-3 R42-R45). PCHG, FUSE, DCHG, DDSG unused, left floating (p75).

PACK_S R18 5.1k R19 5.1k VSYS
LD_S R20 5.1k R21 5.1k VSYS
RST_SHUT R22 1k VSS
TS2_F R23 10k TS2_WAKE
NTC1 C20 1nF VSS
NTC2 C21 1nF VSS
NTC3 C22 1nF VSS

Four thermistors: TS1, TS3, CFETOFF (NTC1-3), HDQ (board).
TS2 stays free as SHUTDOWN wake (p22 pin table, section 13):
switch to VSS through 10k (Fig 16-2 R27); 1nF: p65 limit <4nF.
REGIN, BREG tied to VSS, REG1, REG2 open (p75): I2C pull-ups on LP.
CFETOFF as NTC needs DFETOFF set as BOTHOFF to stop CHG by pin (p48).

SIZING (shunt 0.3 mOhm: no 0.25 mOhm 4-terminal part exists: datasheet p22 steps)
OCD1 24 mV = 80 A, 66.7-93.3 A at +/-4 mV, delay 100 ms OCD3 60 A for 1 s (current-based, inside the AFE, the agreed 60 A ceiling)
OCC 12 mV = 40 A, 33.3-48.8 A at -2/+2.65 mV, under the 50 A limit SCD 60 mV = 200 A, 170-230 A, delay 30 us: bank A short protection (105-165 A, about 10 us) acts first
OCD and OCC are set in 2 mV steps. V(OC_ACC) is -2/+2.65 mV below 20 mV, +/-4 mV 20-56 mV, +/-5 mV 56-100 mV. RS1 takes 6.75 W at 150 A: 5 W part, 100 ms path only.

OPEN ON THIS SHEET
1. Balance FETs are captured below (SLUAA81 Fig 2-2). OTP/data memory owed: Cell Balance Min Cell V 3900 mV, Max Cells 4.
2. CP1 raised to 1 uF for the 4+4 FET gate load (p65 rule): worst case 1.69 uC droops 11 V by 1.7 V, pump start about 130 ms.
3. OTP owed: DFETOFF Pin Config 0x42 (active high, BOTHOFF), CFETOFF and HDQ as thermistor inputs, Presence and DFETOFF logic are on HP_power_board.
   Cell 1 interconnect stays 0: VC0 reads the Kelvin B0 tab, VC0 sits below VSS by the 10 AWG return drop, about 90 mV at 60 A (abs max VSS - 0.3 V).
4. RS1 land: CSS4J-4026 on the BVR4026 library footprint, check against the Bourns drawing.

External balance, SLUAA81A Fig 2-2: internal FET current through VC(n-1) input resistor gives Vgs. 3.9 V: 1.59 V min (RCB 46 R). SSM3K56FS is specified at 1.5 V for 50 mA only and 1.8 V for 200 mA: check Rds at 150 mA on the bench, top cell worst case 2.35 V.
Rbal 27R: 140-150 mA external at 3.9-4.2 V, plus 17-19 mA internal. Cell Balance Max Cells 4: about 2.5 W total.

Q1 R25 BG1 27R 1W 2512 B1 BD1 B0 R26 1k VC0 D16 BZT52C5V1 B0 SSM3K56FS
Q2 R27 BG2 27R 1W 2512 B2 BD2 B1 R28 1k VC1 D17 BZT52C5V1 B1 SSM3K56FS
Q3 R29 BG3 27R 1W 2512 B3 BD3 B2 R30 1k VC2 D18 BZT52C5V1 B2 SSM3K56FS
Q4 R31 BG4 27R 1W 2512 B4 BD4 B3 R32 1k VC3 D19 BZT52C5V1 B3 SSM3K56FS
Q5 R33 BG5 27R 1W 2512 B5 BD5 B4 R34 1k VC4 D20 BZT52C5V1 B4 SSM3K56FS
Q6 R35 BG6 27R 1W 2512 B6 BD6 B5 R36 1k VC5 D21 BZT52C5V1 B5 SSM3K56FS
Q7 R37 BG7 27R 1W 2512 B7 BD7 B6 R38 1k VC6 D22 BZT52C5V1 B6 SSM3K56FS
Q8 R39 BG8 27R 1W 2512 B8 BD8 B7 R40 1k VC7 D23 BZT52C5V1 B7 SSM3K56FS
Q9 R41 BG9 27R 1W 2512 B9 BD9 B8 R42 1k VC8 D24 BZT52C5V1 B8 SSM3K56FS
Q10 R43 BG10 27R 1W 2512 B10 BD10 B9 R44 1k VC9 D25 BZT52C5V1 B9 SSM3K56FS
Q11 R45 BG11 27R 1W 2512 B11 BD11 B10 R46 1k VC10 D26 BZT52C5V1 B10 SSM3K56FS
Q12 R47 BG12 27R 1W 2512 B12 BD12 B11 R48 1k VC11 D27 BZT52C5V1 B11 SSM3K56FS
Q13 R49 BG13 27R 1W 2512 B13 BD13 B12 R50 1k VC12 D28 BZT52C5V1 B12 SSM3K56FS

Pack terminals, sense header, shunt, BQ76952 AFE. Part numbers verify. TI SLUSE13B.

marengo		
Sheet: /HP_power_board/pack_input_shunt_afe/		
File: pack_input_shunt_afe.kicad_sch		
Title: pack_input_shunt_afe		
Size: A2	Date: 2026-11-06	Rev: A-wip
KiCad E.D.A. 10.0.8		Id: 3/27

PDSG FET bypasses DSG only (Fig 16-1): source at CD. Driver sinks 30 uA, so gate-source >= 1M (p45).

R210 1M
Q209 DMP10H400SE
PDSG
PDSG_R

R209
A
B
TAB
PDSG_R
DSG_OUT
GND
100R PWR263S-35-1000F

Common drain (CD): Fig 16-3 pin numbering, gates referenced to the outer sources. Body diodes oppose each other.

CD
Q201 Q202 Q203 Q204 Q205 Q206 Q207 Q208
CHG_G R201 4.7R
CHG_G R202 4.7R
CHG_G R203 4.7R
CHG_G R204 4.7R
DSG_G R205 4.7R
DSG_G R206 4.7R
DSG_G R207 4.7R
DSG_G R208 4.7R
IPT015N10N5
PACK_TOP
DSG_OUT
R5201 0.3mR CSS4J-4026 5W
VSYS_SENS_P
VSYS_SENS_N
TP201 VSYS
VSYS
D207 SMDJ58A GND
C208 100nF 100V
C209 100nF 100V 100uF 80V HHXF800GRA101MJH0G
C210 GND

CHG bank: sources on the stack

DSG bank: sources on the output

CHG drive, Fig 16-3 R37/R31/D5.

CHG_GATED R211 5.1k CHG_G
R212 10M
D201 MMSZ5246B 16V
PACK_TOP

Fig 16-3 C28 C29: across the FETs, two in series.
C201 100nF 100V
C202 100nF 100V

Stack spike: 4 x 100 uF hybrid (right, below U201), 2 uH, 600 A, 90 mOhm pack; peak about 77 V at -20% C. Pack fuse covers a shorted part here. TVS is a backstop (conducts from 64.4 V).
D206 SMDJ58A GND
+3V3_P
U201 INA228AIDGSR
VSYS
Vbus VS A1 A0 SDA SCL Vin+ Vin- GND Alert
I2C_HP_SCL
I2C_HP_SDA
+3V3_HP
C207 100nF
GND

Fig 16-3 Q4 D7 R33 R35 D8: If the output is yanked below GND, Q210 holds the DSG gate at its source.

DSG drive. On: 10.2k (Fig 16-3 R39+R41). Off: PNP Q211 local turn-off, base through 220R in series + diode to the DSG pin, 10k base-emitter, 15R in the collector. Not the 7.5k of Fig 16-3 (see note 2).

DSG_GATED R213 5.1k R214 5.1k DSG_G
D202 BAV19W R215 220R DSG_OFF_B
R217 10M D203 MMSZ5246B 16V
R219 10k D204 MMSZ5246B 16V
D205 BAV19W GND
R216 10k
R218 10M
Q210 BSS123
Q211 MMBT2907A
R220 15R
DSG_OUT
DSG_G

Main path meter, PLAN 3.9. Address 0x4C: A1 to SCL, A0 to GND. +/-40.96 mV range = +/-136.5 A. Bank A short sense reads this shu

C203 C204 C205 C206 100uF 80V HHXF800GRA101MJH0G

Charge input, common port: the charger joins on VSYS, DSG body diode side, as the BQ76952 expects (PACK pin sees it, p45). Also wakes the AFE from ship mode through LD.
PLAN 3.7 says reverse-polarity FET. With a reversed charger the blocker sees 54.6 + 54.6 = 109 V: above the 100 V FET and the LM74700 (75 V, SNOSD17G). So a 150 V Schottky: Vishay V10PM15, pads 1, 2 anode, tab K, 0.6 V, 2.4 W at 4 A. SEE RATIONALE.log
CHARGER_PRESENT: CHG_IN only, so VSYS cannot fake it: high above 34 V, 3.2 V at 54.6 V.

SIZING (design/README 2.2)
Conduction, 80 A: 2 x 0.6 mOhm hot x 80^2 = 7.7 W total. At 10 to 30 A: 0.1 to 1.1 W for these two banks.
FET: IPT015N10N5 Rds 1.5 mOhm max at Vgs 10 V, Vgs +/-20 V, so 11 V drive and 16 V zeners are in range (p69 needs >= 15 V).
Shunt voltage rule p69: 13S x 5 V = 65 V DC, 130 V peak. 100 V FETs meet DC. Peak: both TVS parts hold it under 100 V.

J201 CHG_IN
J202 GND
D208 SMDJ58CA
D209 BAV19W
R220 1M
D210 MMSZ4685 3.6V
R221 62k
C211 1nF
CHARGER_PRESENT
D220 V10PM15-M3/I 150V 10A
A1 A2 K CHG_IN CHG_IN VSYS

SOA must cover 600 A prospective for about 50 us. Reworked after gate 3:
1. Prospective current: 600 A / 4 = 150 A per FET. On-state part: 0.6 mOhm x 600 A = 0.36 V, 216 W bank, nothing.
2. Turn-off: TI gives only 2 us typ, 20 us max at 20 nF for the DSG pin, pulsed, stopping at LD (section 16.5). Ours is 48 nF, so a bare 220R is unsupported.
   Q211: base through 220R and D202, about 45 mA, 10k base-emitter, 15R in the collector sets the peak: 0.62 A from 11 V, under the 800 mA rating. All gate charge 844 nC max in about 2 us. 10 us SOA at 54 V is about 300 A per FE
3. Spike: p72 warns fast turn-off raises the top of stack. 4 x 100 uF hybrid, pack EMF behind 2 uH and 90 mOhm (PLAN pack R): peak about 77 V at -20% C. FET stays under 100 V without avalanche.

CP1 is 1 uF on sheet 1: 8 x 211 nC max = 1.69 uC at p65's rule, droop 1.7 V from 11 V.
PDSG not PCHG: p45 PDSG charges the LOAD before DSG closes; PCHG charges a LOW BATTERY. README 2.1 described the PDSG job.
100R: peak 54.6^2/100 = 29.8 W. VSYS load 1.46 mF (hold-up via Q1101 body diode, plus VSYS bulk): 2.2 J per power-up, tau 0.146 s. PWR263S chart: 10 J at 100 ms.
Set OTP: Mfg Status Init[FET_EN] = 1 (default 0 is FET test mode: no PDSG, VSYS stays down, the MCU never boots). PREDISCHARGE on, Prediscarge Timeout 100 (1.0 s), Stop Delta 2500 mV.
Stop Delta needs under 25 mA, so the timeout must exist: a hard VSYS short ends in SCD.
Enabled Protections C[SCDL] = 1, SCDL Latch Limit 2, Counter Dec Delay 60 s: a persistent short costs R209 30 J about every 65 s, not every 6 s.
Firmware reads FET Status first and holds BUCK12_OFF high only while DSG is off, at every predischarge. It never asserts it with DSG on, so an MCU reset keeps compute up.

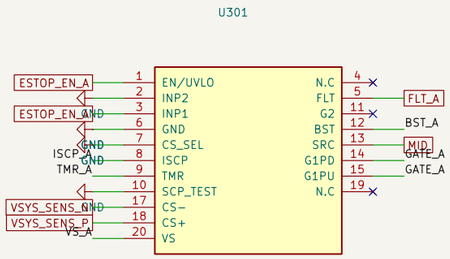
Fig 16-1, Fig 16-3 of TI SLUSE13B. High side, common drain, common port.

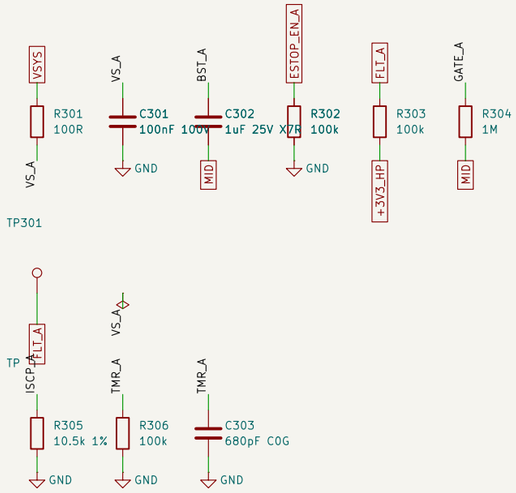
marengo

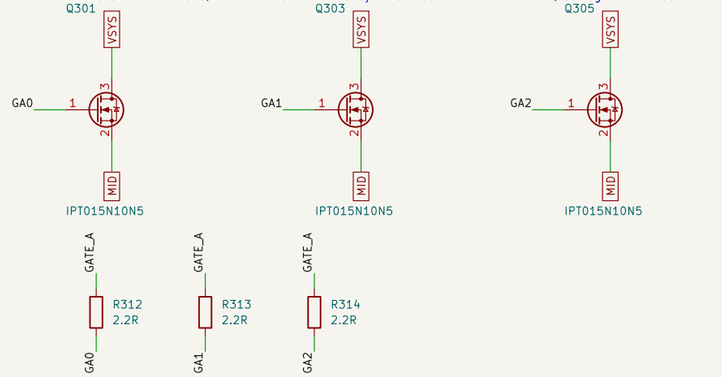
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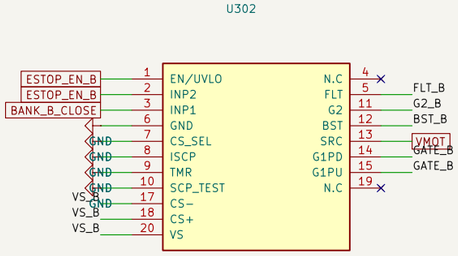
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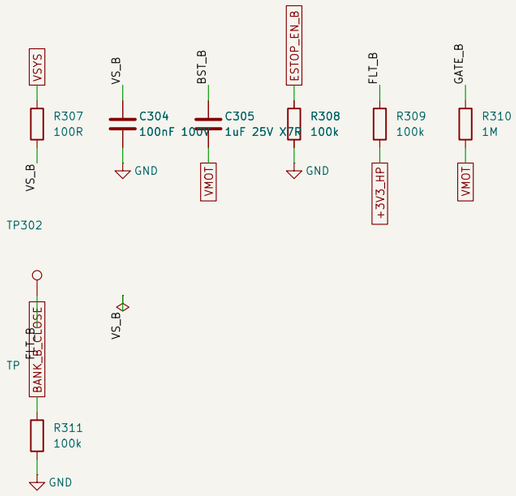
Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 4/27

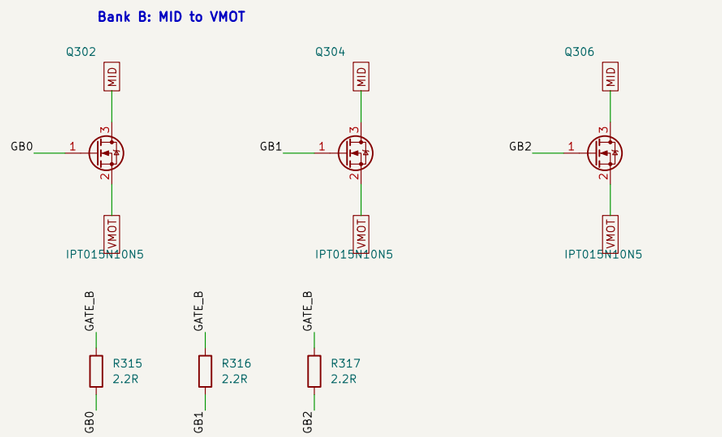








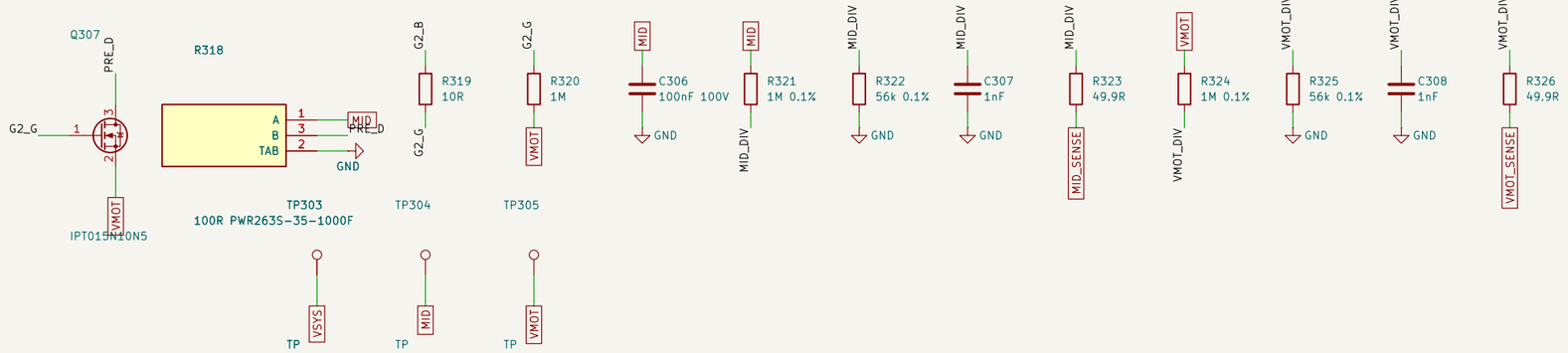




VMOT precharge (IPT015N10N5 SOA is 0.7 A at 54 V, 10 ms, so a hard close on 1.8 mF is out). U302 EN and INP2 on ESTOP_EN_B: loop closed and latch armed, G2 precharges at once, bank B stays open.
BANK_B_CLOSE (INP1, 100k down) closes bank B on command once VMOT reads within 1.5 V of VSYS. Up to 9 mA of VMOT standing load sets a 0.9 V floor across R318, so 0.5 V is out of reach. 0.1% dividers: true gap 1.2 to 1.8 V. 0.81 s, 1.12 s worst case, firmware timeout 1.5 s, then open the loop. EN low kills G1 and G2. Energy 2

Status (TPS48100 has no VDS output): MID, VMOT dividers 1M / 56k, 3.3 V = 62 V, plus VSYS on LP. Bank A short reads MID near VSYS with both open and VMOT low.

Bank A short protection: ISCP 10.5k = 40 mV (31.6–49.5 mV) on RS201 0.3 mOhm = 105–165 A, under BMS SCD and over OCD1 max 93 A. About 10 us, latch-off (100k + 680 pF across TMR). FLT_A to t



TPS48100-Q1 simple-driver mode (SLVSGC0 Fig 7-11). No firmware in the enable path.

marengo

Sheet: /HP_power_board/estop_banks/
File: estop_banks.kicad_sch

Title: estop_banks: E-stop banks A and B, VMOT precharge

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 5/27

Bulk: 18 x 100 uF 80 V hybrid, 1.8 mF. 220 uF 80 V 10x12.5 does not exist. Can is 10 x 16.5 mm: footprint is the closest library match, real land owed.

C401 100uF 80V HHXF800GRA101MJH0G
C402 100uF 80V HHXF800GRA101MJH0G
C403 100uF 80V HHXF800GRA101MJH0G
C404 100uF 80V HHXF800GRA101MJH0G
C405 100uF 80V HHXF800GRA101MJH0G
C406 100uF 80V HHXF800GRA101MJH0G
C407 100uF 80V HHXF800GRA101MJH0G
C408 100uF 80V HHXF800GRA101MJH0G
C409 100uF 80V HHXF800GRA101MJH0G
C410 100uF 80V HHXF800GRA101MJH0G
C411 100uF 80V HHXF800GRA101MJH0G
C412 100uF 80V HHXF800GRA101MJH0G
C413 100uF 80V HHXF800GRA101MJH0G
C414 100uF 80V HHXF800GRA101MJH0G
C415 100uF 80V HHXF800GRA101MJH0G
C416 100uF 80V HHXF800GRA101MJH0G
C417 100uF 80V HHXF800GRA101MJH0G
C418 100uF 80V HHXF800GRA101MJH0G
D401 SMDJ58A
D402 SMDJ58A
C419 1uF 100V
C420 4.7uF 25V
R401 464k
R402 49.9k
R403 9.09k
U403 LM4040AIM3-2.5
U401 TPS7A4001DGNR
R404 499k 0.1%
R405 604k 0.1%
R406 51.1k 0.1%
R407 19.1M
C421 100pF C0G
C422 100nF
Q402 BSS123
J401
J402 M4 terminal, Arcol HS25 10R +
J403 brake R NTC
M4 terminal, Arcol HS25 10R −
U402 TLV1805DBVR
C423 1uF 25V
R408 4.7R
R409 100k
U404 UCC27517DBVR
Q401 IPT015N10N5
RS401 2mR CSS4J-4026K-2L00F 6W
D403 MBRB30H100CT-E3/81
R410 10R
R411 10R
C424 100nF
R412 100R
C425 100nF
U405 INA228AIDGSR

Clamp: 56.77 V on, 55.77 V off (VOH 11.7 V, 19.1M hysteresis). Spread 56.08–57.46 V full temperature. Arcol HS25 10 Ohm 25 W clad, external on 2 mm plate: 322 W, 5.7 A.
FET, 2 mOhm shunt (INA228 SHUNT_CAL 2000, ADCRANGE 0), 30 A Schottky and M4 terminals are sized for the leg stage 2 Ohm resistor, 29 A, 1.7 kW short time: only the external resistor swaps.
INA228 0x4D: clamp power and energy on its own accumulator. Budget about 5 mA, 0.23 W in the LDO.
Failure: stuck-on 322 W: NC thermostat on the HS25 plate sits in the E-stop loop, so it opens the banks in hardware, supervisor hung or not. NTC warns first. SMDJ58A stays off under 64.4 V, spike protection only.
Regen decision (coordinator 11-06): 10 Ohm in rev A. 30 A regen against 5.75 A clamp: BMS opening mid-brake rises 13.5 V/ms, drives stop at 60 V in about 185 us, torque drops. Leg stage fits 2 Ohm.

Regen clamp powers itself from VMOT, works with supervisor, BMS and pack all off (PLAN 3.4).

marengo	
Sheet: /HP_power_board/vmot_bulk_clamp/	
File: vmot_bulk_clamp.kicad_sch	
Title: vmot_bulk_clamp: VMOT bulk, TVS, self-powered regen clamp	
Size: A2	Date: 2026-11-06
KiCad E.D.A. 10.0.8	Rev: A-wip
	Id: 6/27

RSS01
VMOT
SH_P
SH_N
Q_D
3mR WSLP39213L000FEA 5W

Q501
G1
V_OUT
Q_D
IPB017N10N5LFATMA1

Q502
G2
IPB017N10N5LFATMA1

GATE
R501
10R
G1
GATE
R502
10R
G2
V_OUT
D501
MBRB5100T3G
GND
VMOT
C501
1uF 100V
GND
SH_P
C502
100nF 100V
GND
V_OUT
F501
20A 58V 09RT020.WXN
V_OUT
F502
Do not populate, for SMAJ58CA
CANH
D502
SMAJ58CA
GND
CANL
D503
SMAJ58CA
GND
CAN_REF
F503
0.14A MF-MSMF014/60X
GND
TP501
TP
SH_P

J501
V_PORT
J502
M4 terminal, port +
GND
M4 terminal, port –
J503
CANH
CANL
CAN_REF
CAN to trunk

U501
SH_P
SH_N
GATE
V_OUT
VIN
SENSE
GATE
OUT
OVLO
UVLO
PGD
PWR
TIMER
GND
LM5069MMX-1

VMOT
R503
191k
UVLO
UVLO
R504
18.7k
GND
VMOT
R505
95.3k
OVLO
OVLO
R506
4.12k
GND
PWR
R507
69.8k
GND
TIMER
C503
100nF C0G
GND
CDV
D504
5V19W
GATE
CDV
C504
47nF 100V
GND
GATE
R508
C0G1k
CDV_B

Q503
UVLO
EN_INV
GND
BSS123

Q504
EN_INV
PORT_EN
EN_S
BSS123

Q505
GND
CDV_B
CDV
MMBTA92 PNP

VMOT
R509
1M
EN_INV
EN_INV
D505
MMSZ5242B 12V
GND
PORT_EN
R510
100k
GND
VCLAMP12
R511
1M
GND

Q506
EN_S
VCLAMP12
GND
BSS123

U502
+3V3_HP
VB_F
Vbus
VS
A1
A0
SDA
SCL
I2C_HP_SDA
I2C_HP_SCL
IN_P
IN_N
Vin+
Vin–
GND
Alert
INA228AIDGSR

SH_P
R512
10R
IN_P
SH_N
R513
10R
IN_N
IN_P
C505
100nF
IN_N
CV_PORT
R514
100R
VB_F
+3V3_HP
C506
100nF
GND

U503
+3V3_HP
SH_P
SH_N
+IN
–IN
VS
GND
OUT
IM_RAW
INA241A3IDDFR

IM_RAW
R515
49.9R
IM_RAW
+3V3_HP
C507
100nF
GND

INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/-48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1–15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.

Load-side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

U504
TL_R
GND
TL_K
TL431BIDBZR

Q507
CAS_D
CAS_G
TL_K
BSS123

Q508
PG
G
D
S
PD
V_PORT
DMP10H400SE-13

R523
A
B
TAB
PD
GND
GND
150R PWR263S-35-150F

V_PORT
R516
25.5k 0.1%
TL_M
TL_M
R517
34.0k 0.1%
TL_R
TL_R
R518
2.61k 0.1%
GND
PD
R519
3.48M
TL_R
V_PORT
R520
1M
CAS_G
CAS_G
D506
MMSZ5246B
GND
CAS_D
R521
33k
PG
V_PORT
R522
10k
PG
V_PORT
D507
MMSZ5246B
PG

Values shown: arm, waist, aux. LM5069 limit 16.2–20.5 A (48.5–61.5 mV / 3 mOhm), breaker 26.7–43.3 A, RPWR 69.8k = 200 W, fault timer 2.8–9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5–61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501, 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, EN_INV is pulled from VMOT itself with a 12 V zener, and Q3 sinks it through a FET gated by VCLAMP12, so a lost clamp supply turns the port off. PORT_EN comes through a tri-state buffer: E-stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non-isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/-58 V, differential +/-45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/-58 V clamp. Capacitance unchecked: scope on the bus.

CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAK. 56 + 56 + 0.7–15 = 113–127 Ohm. Jumper in for the end node.

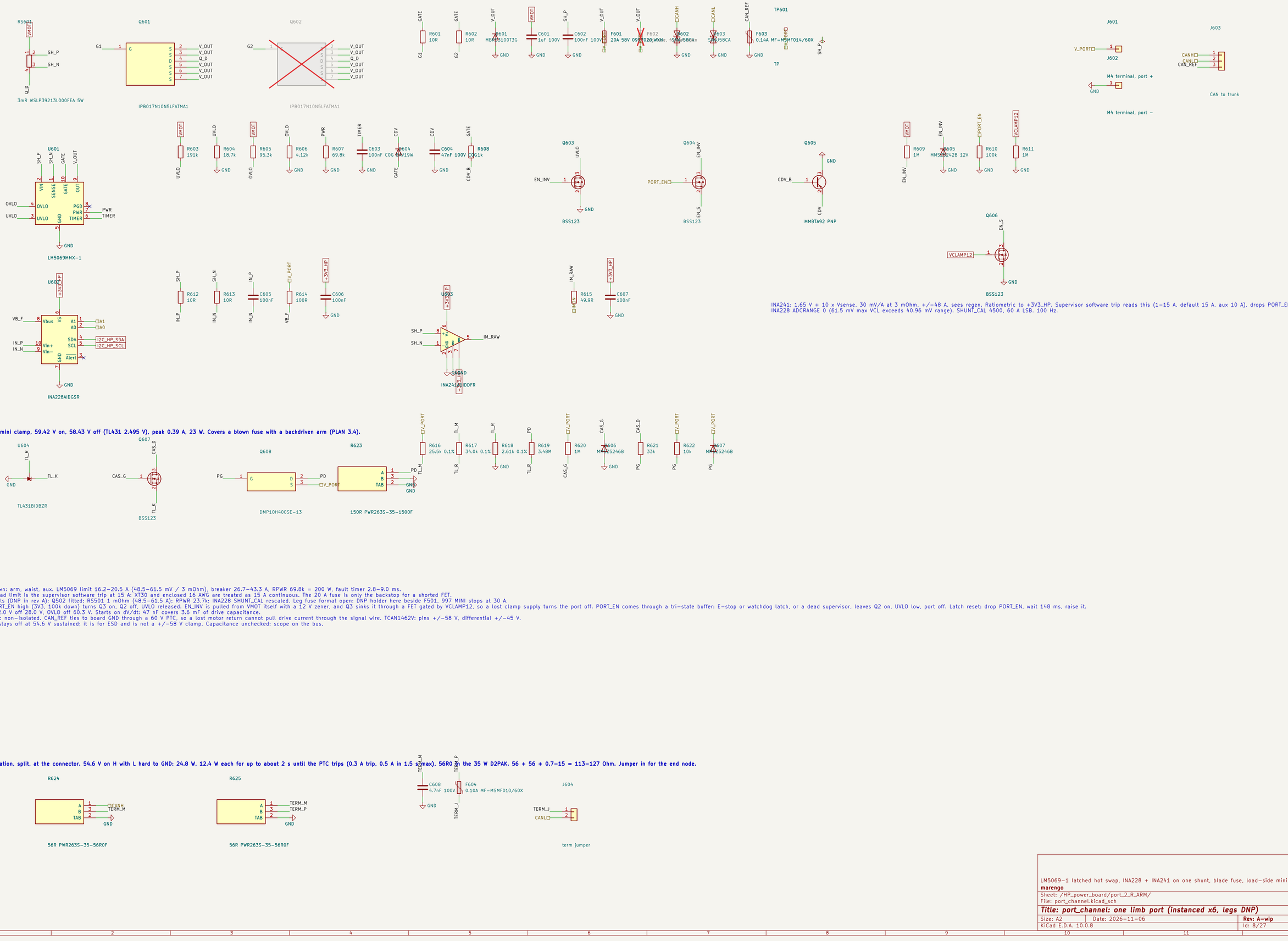
R524
A
B
TAB
CANH
TERM_M
GND
56R PWR263S-35-56R0F

R525
A
B
TAB
TERM_M
TERM_P
GND
56R PWR263S-35-56R0F

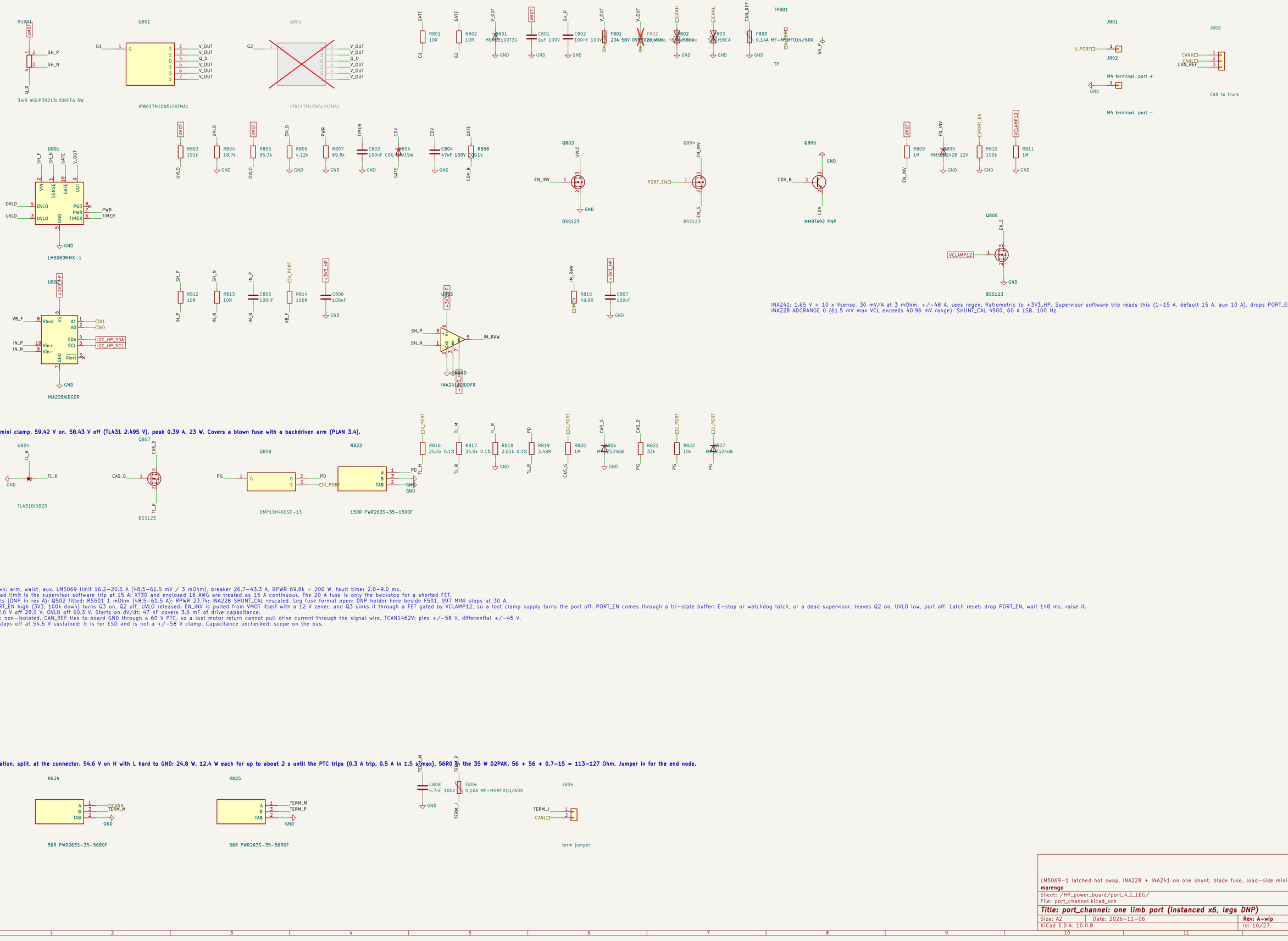
TERM_M
C508
4.7nF 100V
GND
TERM_P
F504
0.10A MF-MSMF010/60X
TERM_J

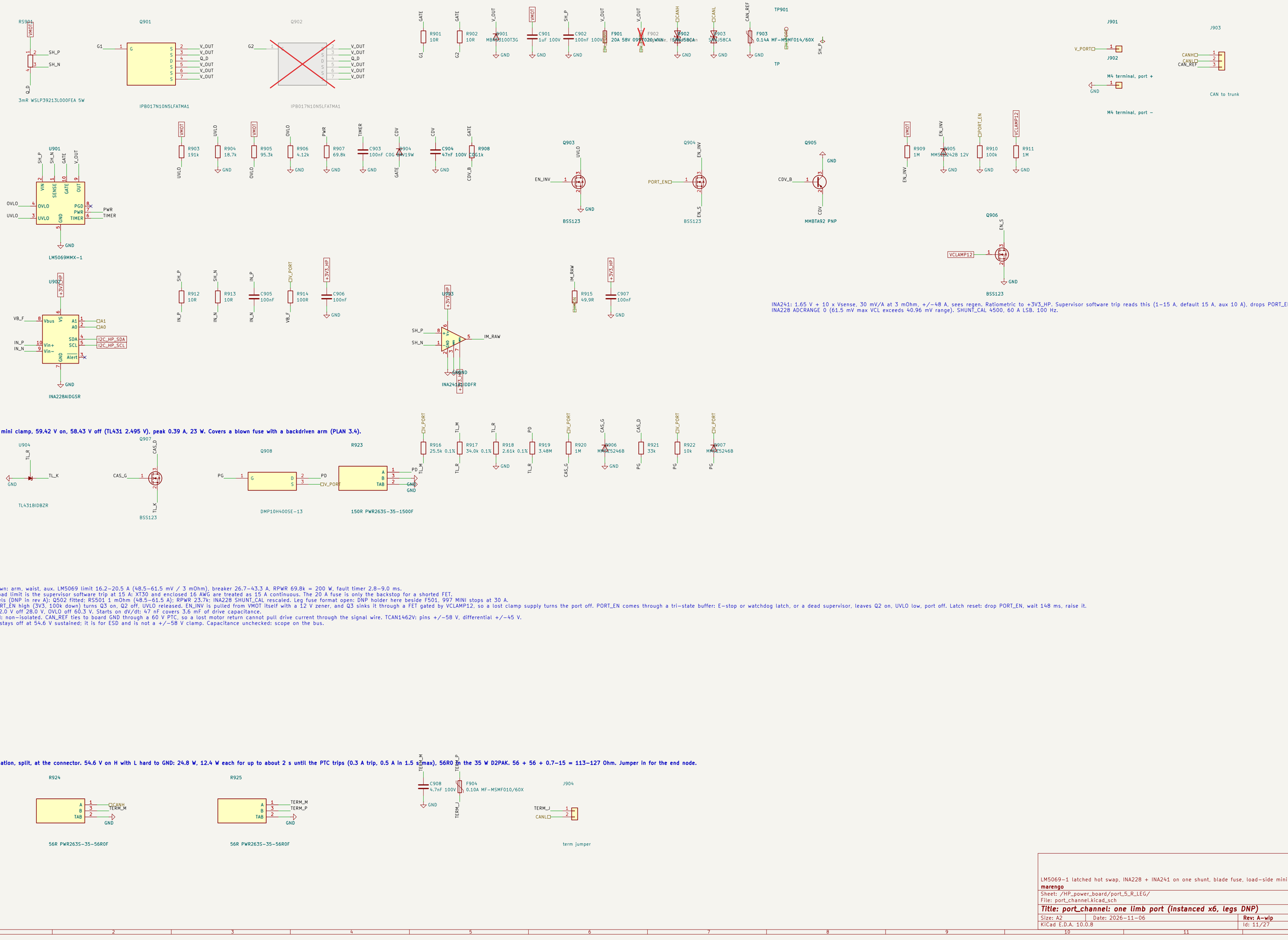
J504
TERM_J
CANL
term jumper

LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.
marengo
Sheet: /HP_power_board/port_1_L_ARM/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2 | Date: 2026-11-06 | **Rev: A-wip**
KiCad E.D.A. 10.0.8 | Id: 7/27

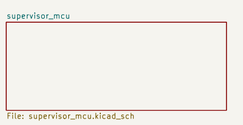

RS601
VMOT
SH_P
SH_N
Q_D
3mR WSLP39213L000FEA 5W
Q601
G1
V_OUT
Q_D
IPB017N10N5LFATMA1
Q602
G2
IPB017N10N5LFATMA1
GATE
R601 10R
R602 10R
V_OUT
MBRS5100T3G
VMOT
C601 1uF 100V
SH_P
C602 100nF 100V
F601 20A 58V 0R0T020.WXN
F602 Do not populate, for testing only
CANH
SMAJ58CA
CANL
CAN_REF
F603 0.14A MF-MSMF014/60X
TP601
TP
J601
V_PORT
J602
M4 terminal, port +
GND
M4 terminal, port −
J603
CANH
CANL
CAN_REF
CAN to trunk
U601
SH_P
SH_N
GATE
V_OUT
VIN
SENSE
OUT
OVLO
UVLO
PGD
PWR
TIMER
LM5069MMX-1
R603 191k
R604 18.7k
R605 95.3k
R606 4.12k
R607 69.8k
C603 100nF C0G
C604 47nF 100V
R608 C0G1k
CDV
C604 47nF 100V
CDV_B
Q603
EN_INV
BSS123
Q604
PORT_END
EN_S
Q605
MMBTA92 PNP
R609 1M
R605 MMSZ4242B 12V
PORT_EN
R610 100k
VCLAMP12
R611 1M
Q606
BSS123
U602
+3V3_HP
VB_F
A1
A0
I2C_HP_SDA
I2C_HP_SCL
IN_P
IN_N
Vbus
Vin+
Vin−
Alert
SDA
SCL
INA228AIDGSR
R612 10R
R613 10R
C605 100nF
R614 100R
C606 100nF
U603
IM_RAW
R615 49.9R
C607 100nF
INA241A3IDDFR
INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1–15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.
Load-side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).
U604
TL_R
TL_K
TL431BIDBZR
Q607
CAS_D
CAS_G
BSS123
Q608
PG
PD
V_PORT
DMP10H400SE-13
R623
150R PWR263S-35-150F
R616 25.5k 0.1%
R617 34.0k 0.1%
TL_M
R618 2.61k 0.1%
R619 3.48M
PD
R620 1M
R606 MMSZ5246B
R621 33k
R622 10k
R607 MMSZ5246B
Values shown: arm, waist, aux. LM5069 limit 16.2–20.5 A (48.5–61.5 mV / 3 mOhm), breaker 26.7–43.3 A, RPWR 69.8k = 200 W, fault timer 2.8–9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5–61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501, 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. EN_INV is pulled from VMOT itself with a 12 V zener, and Q3 sinks it through a FET gated by VCLAMP12, so a lost clamp supply turns the port off. PORT_EN comes through a tri-state buffer: E-stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non-isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/−58 V, differential +/−45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/−58 V clamp. Capacitance unchecked: scope on the bus.
CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each on the 35 W D2PAK. 56 + 56 + 0.7–15 = 113–127 Ohm. Jumper in for the end node.
R624
CANH
TERM_M
56R PWR263S-35-56R0F
R625
TERM_P
TERM_M
C608 4.7nF 100V
F604 0.10A MF-MSMF010/60X
J604
TERM_J
CANL
term jumper
LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.
marengo
Sheet: /HP_power_board/port_2_R_ARM/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2
Date: 2026-11-06
Rev: A-wip
KiCad E.D.A. 10.0.8
Id: 8/27

RS701
VMOT
SH_P
SH_N
Q_D
3mR WSLP39213L000FEA 5W
Q701
G1
V_OUT
Q_D
IPB017N10N5LFATMA1
Q702
G2
IPB017N10N5LFATMA1
GATE
R701 10R
R702 10R
V_OUT
D701 MBRS3100T3G
VMOT
C701 1uF 100V
SH_P
C702 100nF 100V
F701 20A 58V 09R7020.WXN
F702 Do not place, for 58V use 5BCA
CANH
SMAJ58CA
CANL
CAN_REF
F703 0.14A MF-MSMF014/60X
TP701
TP
SH_P
J701
V_PORT
J702
M4 terminal, port +
GND
M4 terminal, port −
J703
CANH
CANL
CAN_REF
CAN to trunk
U701
SH_P
SH_N
GATE
V_OUT
VIN
SENSE
OUT
OVLO
UVLO
GND
PGD
PWR
TIMER
LM5069MMX-1
R703 191k
R704 18.7k
R705 95.3k
R706 4.12k
R707 69.8k
C703 100nF C0G
D704 PTVS19W
C704 47nF 100V
R708 C0G1k
CDV
CDV_B
Q703
EN_INV
BSS123
Q704
PORT_END
EN_S
Q705
MMBTA92 PNP
R709 1M
MMSZ5242B 12V
R710 100k
R711 1M
PORT_EN
VCLAMP12
Q706
U702
+3V3_HP
VB_F
IN_P
IN_N
Vbus
VS
A1
A0
SDA
SCL
Vin+
Vin−
Alert
I2C_HP_SDA
I2C_HP_SCL
INA228AIDGSR
R712 10R
R713 10R
C705 100nF
R714 100R
C706 100nF
CV_PORT
U703
IM_RAW
INA241B1IDDFR
R715 49.9R
C707 100nF
INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1−15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.
Load−side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).
U704
TL_R
TL_K
TL431BIDBZR
Q707
CAS_D
CAS_G
BSS123
Q708
PG
PD
V_PORT
DMP10H400SE-13
R723
150R PWR263S-35-150F
R716 25.5k 0.1%
R717 34.0k 0.1%
R718 2.61k 0.1%
R719 3.48M
R720 1M
D706 MMSZ5246B
R721 33k
R722 10k
D707 MMSZ5246B
Values shown: arm, waist, aux. LM5069 limit 16.2−20.5 A (48.5−61.5 mV / 3 mOhm), breaker 26.7−43.3 A, RPWR 69.8k = 200 W, fault timer 2.8−9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5−61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501, 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. EN_INV is pulled from VMOT itself with a 12 V zener, and Q3 sinks it through a FET gated by VCLAMP12, so a lost clamp supply turns the port off. PORT_EN comes through a tri−state buffer: E−stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non−isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/−58 V, differential +/−45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/−58 V clamp. Capacitance unchecked: scope on the bus.
CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAK. 56 + 56 + 0.7−15 = 113−127 Ohm. Jumper in for the end node.
R724
CANH
TERM_M
56R PWR263S-35-56R0F
R725
TERM_M
TERM_P
56R PWR263S-35-56R0F
TERM_M
TERM_P
C708 4.7nF 100V
F704 0.10A MF-MSMF010/60X
TERM_J
J704
CANL
term jumper
LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.
marengo
Sheet: /HP_power_board/port_3_WAIST/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2
Date: 2026−11−06
Rev: A−wip
KiCad E.D.A. 10.0.8
Id: 9/27

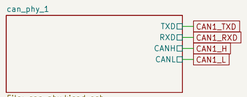

RS801
VMOT
SH_P
SH_N
Q_D
3mR WSLP39213L000FEA 5W
Q801
G1
V_OUT
Q_D
IPB017N10N5LFATMA1
Q802
G2
IPB017N10N5LFATMA1
GATE
R801 10R
R802 10R
V_OUT
MBRB2100T3G
GND
VMOT
C801 1uF 100V
SH_P
C802 100nF 100V
F801 20A 58V 0R0T020.WXN
F802 Do not populate, for 58V use
CANH
SMAJ58CA
CANL
CAN_REF
F803 0.14A MF-MSMF014/60X
TP801
TP
SH_P
J801
V_PORT
J802
M4 terminal, port +
M4 terminal, port −
J803
CANH
CANL
CAN_REF
CAN to trunk
U801
SH_P
SH_N
GATE
V_OUT
VIN
SENSE
OUT
OVLO
UVLO
PGD
PWR
TIMER
LM5069MMX-1
R803 191k
R804 18.7k
R805 95.3k
R806 4.12k
R807 69.8k
C803 100nF C0G
C804 47nF 100V
R808 C0G1k
CDV
C804 V19W
CDV_B
Q803
EN_INV
BSS123
Q804
PORT_EN
EN_S
Q805
MMBTA92 PNP
R809 1M
MMSZ5242B 12V
R810 100k
R811 1M
PORT_EN
VCLAMP12
Q806
U802
+3V3_HP
VB_F
Vbus
VS
A1
A0
SDA
SCL
I2C_HP_SDA
I2C_HP_SCL
IN_P
IN_N
Vin+
Vin−
Alert
INA228AIDGSR
R812 10R
R813 10R
C805 100nF
R814 100R
C806 100nF
U803
IM_RAW
INA241B1IDDFR
R815 49.9R
C807 100nF
INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1−15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.
Load−side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).
U804
TL_R
TL_K
GND
TL431BIDBZR
Q807
CAS_D
CAS_G
BSS123
Q808
PG
PD
V_PORT
DMP10H400SE-13
R823
150R PWR263S-35-150F
R816 25.5k 0.1%
R817 34.0k 0.1%
R818 2.61k 0.1%
R819 3.48M
R820 1M
R806 MMSZ5246B
R821 33k
R822 10k
R807 MMSZ5246B
Values shown: arm, waist, aux. LM5069 limit 16.2−20.5 A (48.5−61.5 mV / 3 mOhm), breaker 26.7−43.3 A, RPWR 69.8k = 200 W, fault timer 2.8−9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5−61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501, 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, EN_INV is pulled from VMOT itself with a 12 V zener, and Q3 sinks it through a FET gated by VCLAMP12, so a lost clamp supply turns the port off. PORT_EN comes through a tri−state buffer: E−stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non−isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/−58 V, differential +/−45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/−58 V clamp. Capacitance unchecked: scope on the bus.
CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAK. 56 + 56 + 0.7−15 = 113−127 Ohm. Jumper in for the end node.
R824
CANH
TERM_M
56R PWR263S-35-56R0F
R825
TERM_P
TERM_M
C808 4.7nF 100V
F804 0.10A MF-MSMF010/60X
J804
TERM_J
CANL
term jumper
LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.
marengo
Sheet: /HP_power_board/port_4_L_LEG/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2
Date: 2026−11−06
Rev: A−wip
KiCad E.D.A. 10.0.8
Id: 10/27

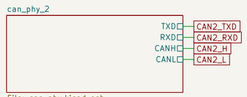

RS901
VMOT
SH_P
SH_N
Q_D
3mR WSLP39213L000FEA 5W
Q901
G1
V_OUT
Q_D
IPB017N10N5LFATMA1
Q902
G2
IPB017N10N5LFATMA1
GATE
R901 10R
R902 10R
V_OUT
D901 MBRS3100T3G
GND
VMOT
C901 1uF 100V
SH_P
C902 100nF 100V
F901 20A 58V 0R1T020.WXN
F902 Do not fit. fuse bypass
CANH
D902 SMAJ58CA
CANL
D903 SMAJ58CA
CAN_REF
F903 0.14A MF-MSMF014/60X
TP901
TP
SH_P
J901
V_PORT
J902
M4 terminal, port +
M4 terminal, port −
J903
CANH
CANL
CAN_REF
CAN to trunk
U901
SH_P
SH_N
GATE
V_OUT
VIN
SENSE
OUT
OVLO
UVLO
PGD
PWR
TIMER
LM5069MMX-1
R903 191k
R904 18.7k
R905 95.3k
R906 4.12k
R907 69.8k
C903 100nF C0G
D904 PTVS19W
CDV
C904 47nF 100V
R908 C0G1k
CDV_B
Q903
EN_INV
BSS123
Q904
PORT_END
EN_S
Q905
MMBTA92 PNP
R909 1M
D905 MMSZ5242B 12V
PORT_EN
R910 100k
VCLAMP12
R911 1M
Q906
U902
+3V3_HP
VB_F
IN_P
IN_N
A1
A0
SDA
SCL
I2C_HP_SDA
I2C_HP_SCL
Alert
INA228AIDGSR
R912 10R
R913 10R
C905 100nF
R914 100R
C906 100nF
U903
IM_RAW
INA241A3IDDFR
R915 49.9R
C907 100nF
INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1−15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.
Load−side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).
U904
TL_R
TL_K
TL431BIDBZR
Q907
CAS_D
CAS_G
BSS123
Q908
PG
PD
V_PORT
DMP10H400SE-13
R923
150R PWR263S-35-150F
R916 25.5k 0.1%
TL_M
R917 34.0k 0.1%
R918 2.61k 0.1%
R919 3.48M
R920 1M
D906 MMSZ5246B
R921 33k
R922 10k
D907 MMSZ5246B
Values shown: arm, waist, aux. LM5069 limit 16.2−20.5 A (48.5−61.5 mV / 3 mOhm), breaker 26.7−43.3 A, RPWR 69.8k = 200 W, fault timer 2.8−9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5−61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501, 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, EN_INV is pulled from VMOT itself with a 12 V zener, and Q3 sinks it through a FET gated by VCLAMP12, so a lost clamp supply turns the port off. PORT_EN comes through a tri−state buffer: E−stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non−isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/−58 V, differential +/−45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/−58 V clamp. Capacitance unchecked: scope on the bus.
CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAP. 56 + 56 + 0.7−15 = 113−127 Ohm. Jumper in for the end node.
R924
CANH
TERM_M
56R PWR263S-35-56R0F
R925
TERM_P
TERM_M
C908 4.7nF 100V
F904 0.10A MF-MSMF010/60X
J904
TERM_J
CANL
term jumper
LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.
marengo
Sheet: /HP_power_board/port_5_R_LEG/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2
Date: 2026−11−06
Rev: A−wip
KiCad E.D.A. 10.0.8
Id: 11/27

RS1001
VMOT
SH_P
SH_N
Q_D
3mR WSLP39213L000FEA 5W
Q1001
G1
V_OUT
Q_D
IPB017N10N5LFATMA1
Q1002
G2
IPB017N10N5LFATMA1
GATE
R1001 10R
R1002 10R
V_OUT
D1001 MBRB2100T3G
GND
C1001 1uF 100V
C1002 100nF 100V
SH_P
F1001 20A 58V 09R7020.WXN
F1002 Do not populate. fuse for testing
CANH
D1002 SMAJ58CA
CANL
D1003 SMAJ58CA
CAN_REF
F1003 0.14A MF-MSMF014/60X
TP1001
TP
SH_P
J1001
V_PORT
J1002
M4 terminal, port +
M4 terminal, port −
J1003
CANH
CANL
CAN_REF
CAN to trunk
U1001
LM5069MMX-1
VIN
SENSE
GATE
OUT
OVLO
UVLO
PGD
PWR
TIMER
R1003 191k
R1004 18.7k
R1005 95.3k
R1006 4.12k
R1007 69.8k
C1003 100nF C0G
D1004 PTV19W
C1004 47nF 100V
R1008 C0G1k
CDV_B
Q1003
BSS123
EN_INV
Q1004
PORT_END
EN_S
Q1005
MMBTA92 PNP
CDV
R1009 1M
D1005 MMSZ5242B 12V
R1010 100k
R1011 1M
PORT_EN
VCLAMP12
Q1006
U1002
INA228AIDGSR
+3V3_HP
VB_F
IN_P
IN_N
I2C_HP_SDA
I2C_HP_SCL
R1012 10R
R1013 10R
C1005 100nF
R1014 100R
C1006 100nF
U1003
INA241A3IDDFR
IM_RAW
R1015 49.9R
C1007 100nF
INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1–15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.
Load-side mini clamp, 59.42 V on, 58.43 V off (TL431 2.495 V), peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).
U1004
TL431BIDBZR
TL_R
TL_K
Q1007
BSS123
CAS_D
CAS_G
Q1008
DMP10H400SE-13
PG
PD
R1023
150R PWR263S-35-150F
R1016 25.5k 0.1%
R1017 34.0k 0.1%
R1018 2.61k 0.1%
R1019 3.48M
R1020 1M
D1006 MMSZ5246B
R1021 33k
R1022 10k
D1007 MMSZ5246B
TL_M
Values shown: arm, waist, aux. LM5069 limit 16.2–20.5 A (48.5–61.5 mV / 3 mOhm), breaker 26.7–43.3 A, RPWR 69.8k = 200 W, fault timer 2.8–9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5–61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501, 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, EN_INV is pulled from VMOT itself with a 12 V zener, and Q3 sinks it through a FET gated by VCLAMP12, so a lost clamp supply turns the port off. PORT_EN comes through a tri-state buffer: E-stop or watchdog latch, or a dead supervisor, leaves Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
Bus ground: non-isolated. CAN_REF ties to board GND through a 60 V PTC, so a lost motor return cannot pull drive current through the signal wire. TCAN1462V: pins +/−58 V, differential +/−45 V.
SMAJ58CA stays off at 54.6 V sustained; it is for ESD and is not a +/−58 V clamp. Capacitance unchecked: scope on the bus.
CAN termination, split, at the connector. 54.6 V on H with L hard to GND: 24.8 W, 12.4 W each for up to about 2 s until the PTC trips (0.3 A trip, 0.5 A in 1.5 s max), 56R0 each in the 35 W D2PAK. 56 + 56 + 0.7–15 = 113–127 Ohm. Jumper in for the end node.
R1024
56R PWR263S-35-56R0F
R1025
TERM_M
TERM_P
C1008 4.7nF 100V
F1004 0.10A MF-MSMF010/60X
J1004
TERM_J
CANL
term jumper
LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.
marengo
Sheet: /HP_power_board/port_6_AUX/
File: port_channel.kicad_sch
Title: port_channel: one limb port (instanced x6, legs DNP)
Size: A2
Date: 2026-11-06
Rev: A-wip
KiCad E.D.A. 10.0.8
Id: 12/27

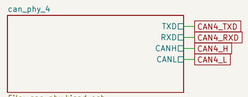
















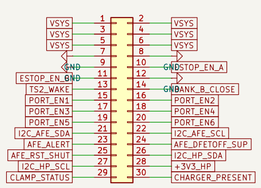


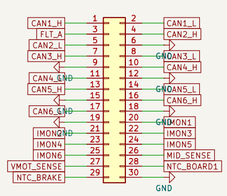




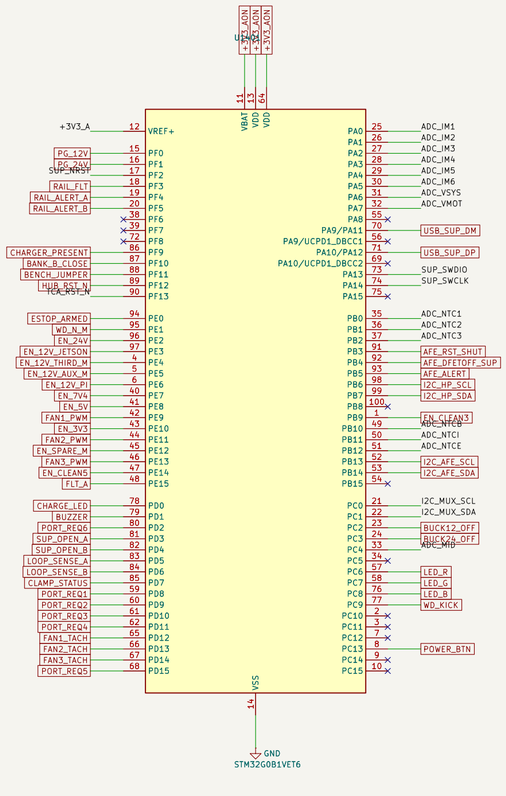


Board-to-board: mirror of HP J101, J102.



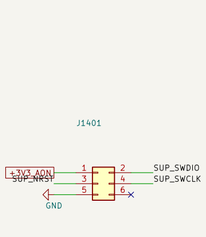


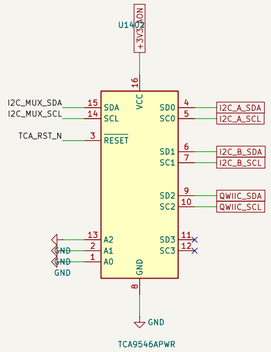
marengo		
Sheet: /LP_logic_board/		
File: LP_logic_board.kicad_sch		
Title: LP_logic_board: low-power board		
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 13/27



G0B1VET6 pin plan: parts check 68 of 68 against ST pin data. Extra lines beyond the plan: PC2/PC3 buck off, PF0/PF1 PG, PF3–5 rail flags, PB9 EN_CLEAN3. PF10 was BOARD_PRESENT, now BANK_B_CLOSE. PC9 WD_KICK, PE15 FLT_A, PE0 ESTOP_ARMED, PE1 WD_N read-back, input only, 10k in series. PD8–11, PD15, PD2 are now PORT_REQ, g
I2C: 3 hardware buses. I2C1 HP meters. I2C2 AFE. I2C3 to TCA9546A at 0x70 (ch0 meters A, ch1 meters B, ch2 Qwiic, ch3 spare).
Five NTC channels sit on 10k to the ADC reference. NTC_BOARD1 also reads open when the HP board is missing: replaces BOARD_PRESENT.

12 V LOAD SHED (budget sums to 22 A on a 12.5 A converter, DC limit 14.2 A min). U1321 meters the real total on +12V: amps are not summed across rails. Start values, tune on the bench.
1. Above 10.5 A, 500 ms: fans to reduced duty (FANx_PWM).
2. Above 11.0 A, 500 ms: third board off (EN_12V_THIRD).
3. HARDWARE, above 12.0 A, about 150 us: U1321 ALERT latches SHED_N: third board, aux and spare off through 1k series FETs, no firmware. Firmware clears after the total stays under 10.5 A.
4. Above 12.4 A for 1 s with step 3 done: Pi stub and Jetson last, after telling the host. PG_12V falling alone never sheds compute: log only.
PORT SOFTWARE TRIP (firmware): each ADC_IMx, 1–15 A per port, default 15 A on arm and waist, 10 A on aux. Over the limit drops PORT_REQx, then reset needs the 148 ms latch sequence.
WATCHDOG: kick WD_KICK from the main loop only, after the port and pack checks ran. 1.36–1.84 s, not firmware-startable or stoppable.





Supervisor can open the banks and ports, never close them against the loop. All enables fail off.

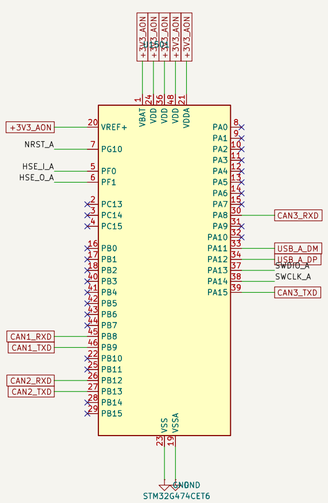
marengo

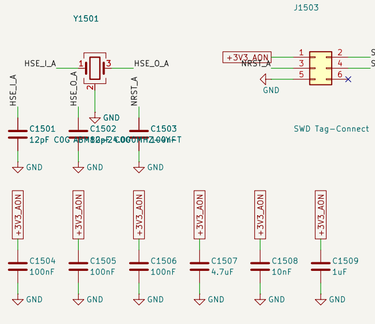
Sheet: /LP_logic_board/supervisor_mcu/
File: supervisor_mcu.kicad_sch

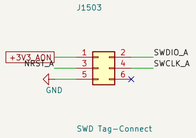
Title: supervisor_mcu: STM32G0B1 supervisor, I2C mux

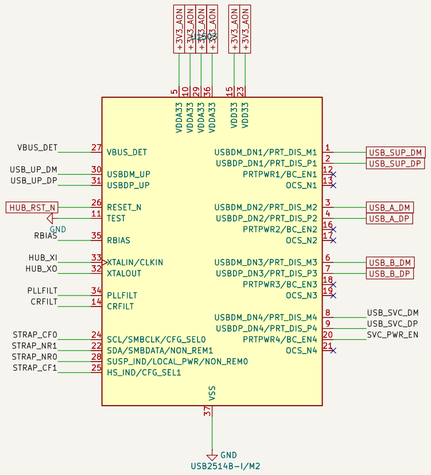
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 14/27

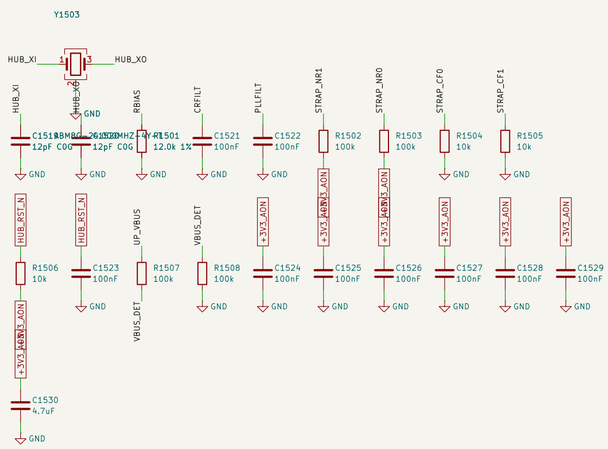
Hub: DN1 supervisor, DN2 bridge A, DN3 bridge B, DN4 service. No EEPROM: CFG_SEL 00, NON_REM 11, self powered, individual switching. TEST to GND. OCS_N1–4 open (internal pull-up).
FDCAN1_RX on PB8-BOOT0: program option bytes to ignore the pin before first run. PA15 released from JTDI for FDCAN3_TX. Firmware: budgetcan_fw, candleLight has no G474 build.
Service USB-C sits on a hub downstream port, so it is a host port: a laptop cannot reach the board through it. Rp 56k, default USB current.

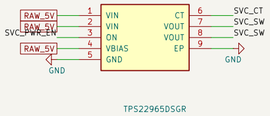


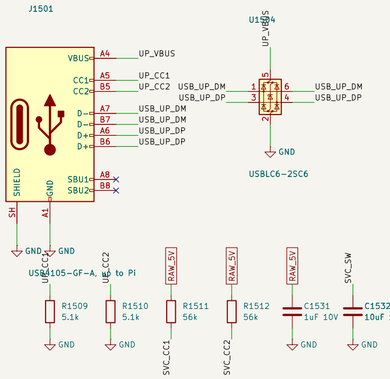


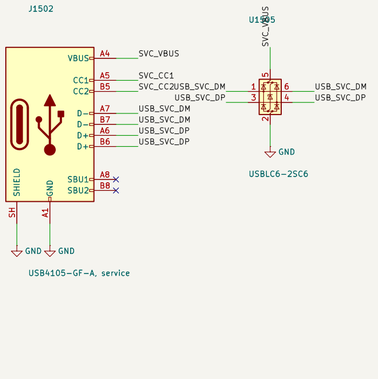












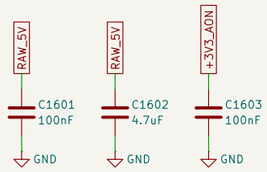
6 CAN buses over one USB-C to the Pi. Hub comes up without the supervisor, so the Pi keeps CAN if

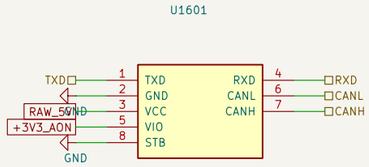
marengo

Sheet: /LP_logic_board/can_bridges_usb_hub/
File: can_bridges_usb_hub.kicad_sch

Title: can_bridges_usb_hub: two STM32G474 CAN bridges, USB2514B hub

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 15/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

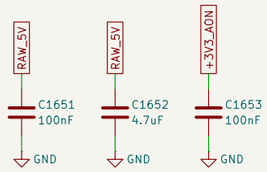
TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

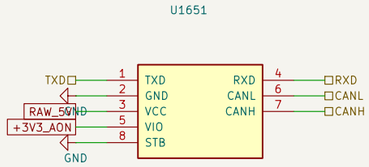
marengo

Sheet: /LP_logic_board/can_phy_1/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 16/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

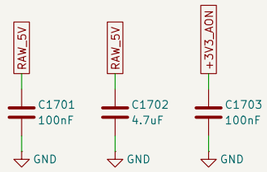
TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

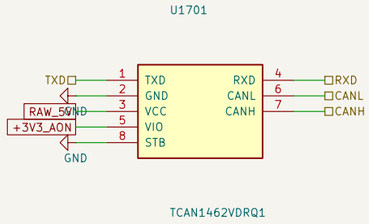
marengo

Sheet: /LP_logic_board/can_phy_2/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 17/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

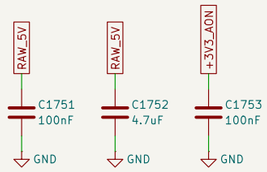
TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

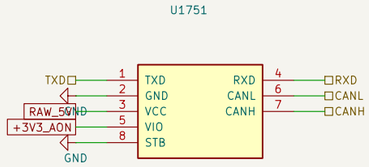
marengo

Sheet: /LP_logic_board/can_phy_3/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 18/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

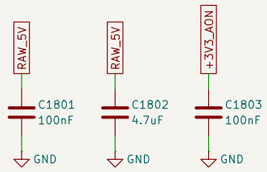
TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

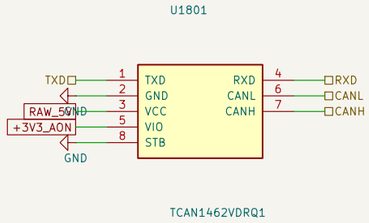
marengo

Sheet: /LP_logic_board/can_phy_4/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 19/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

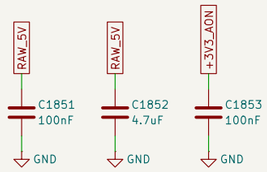
TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

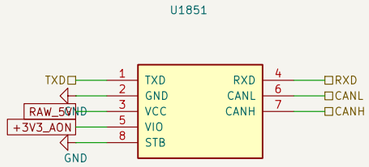
marengo

Sheet: /LP_logic_board/can_phy_5/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 20/27





VCC 5 V, 70 mA dominant max, 130 mA with a bus fault: 0.42 A from RAW_5V for six in dominant. VIO 3.3 V logic. STB to GND: normal mode, it has an internal pull-up.
Bus pins +/-58 V to GND, +/-45 V differential, TXD dominant timeout. Termination and TVS sit at the port connector on HP. Single fault H or L to 54.6 V: the other line follows through the termination, in range.
H at 54.6 V and L hard to GND together exceeds +/-45 V: the transceiver may die, nothing burns. Not isolated: see RATIONALE-log.

TCAN1462V-Q1, non-isolated, +/-58 V bus pins. Motors and frame already tie CAN to power GND.

marengo

Sheet: /LP_logic_board/can_phy_6/
File: can_phy.kicad_sch

Title: can_phy: one CAN transceiver (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 21/27

Ideal diode and hold-up. LM5050-1 not LM74700: PLAN 2.3 wants 75 V; LM74700 stops at 65 V, and VSYS needs no reverse-polarity function here.
20 x 68 uF 80 V hybrid, 10 x 10.8 mm = 1.36 mF, fits the 13 mm board gap: 12.6 ms at 90 W from 46.8 V, 6.8 ms from 37 V. README claimed 1 mF for 10 ms: true only above 48 V.
The 24 V converter taps VSYS ahead of the diode so 75 W does not drain the hold-up.

U1101
LM5050MK-1
Q1101
CSD19532Q5B
C1101 100nF 100V
C1102 100nF 100V
D1101 B340 30V
U1102
LM5145RGYR
Q1102
Q1103
BSC040N10NS5ATMA1
L1101 XAL1510-682MED 33.2k
R1101 33.2k
R1102 33.2k
R1103 2.37k
R1104 6.04k
C1123 10nF
C1124 33pF
R1105 100R
C1125 1nF
C1126 33nF
C1127 100nF 25V
R1106 2.2R
C1128 2.2uF 25V
R1107 348R
C1129 18pF C0G 100V
R1108 1.21M
R1109 49.9k
C1130 100nF 100V
R1110 20k
C1131 4.7uF 100V
C1132 4.7uF 100V
C1133 4.7uF 100V
C1134 4.7uF 100V
C1135 4.7uF 100V
C1136 22uF 25V
C1137 22uF 25V
C1138 22uF 25V
C1139 22uF 25V
C1140 22uF 25V
C1141 22uF 25V
C1142 22uF 25V
Q1104
BSS123
R1111 100k
U1103
LM5145RGYR
Q1105
Q1106
CSD19534Q5A
L1102 XAL1510-333MED 33.2k
R1112 33.2k
R1113 80.6k
R1114 2.8k
R1115 17.8k
C1143 6.8nF
C1144 10pF
R1116 100R
C1145 680pF
C1146 33nF
C1147 100nF 25V
R1117 2.2R
C1148 2.2uF 25V
R1118 301R
C1149 22pF C0G 100V
R1119 499k
R1120 18.2k
C1150 100nF 100V
R1121 20k
C1151 4.7uF 100V
C1152 4.7uF 100V
C1153 4.7uF 100V
C1154 10uF 50V
C1155 10uF 50V
C1156 10uF 50V
C1157 10uF 50V
C1158 10uF 50V
C1159 10uF 50V
C1160 10uF 50V
Hold-up bank, 20 x 68 uF, VSYS_OR to GND
C1103 68uF 80V B40950B0686M100
C1104 68uF 80V B40950B0686M100
C1105 68uF 80V B40950B0686M100
C1106 68uF 80V B40950B0686M100
C1107 68uF 80V B40950B0686M100
C1108 68uF 80V B40950B0686M100
C1109 68uF 80V B40950B0686M100
C1110 68uF 80V B40950B0686M100
C1111 68uF 80V B40950B0686M100
C1112 68uF 80V B40950B0686M100
C1113 68uF 80V B40950B0686M100
C1114 68uF 80V B40950B0686M100
C1115 68uF 80V B40950B0686M100
C1116 68uF 80V B40950B0686M100
C1117 68uF 80V B40950B0686M100
C1118 68uF 80V B40950B0686M100
C1119 68uF 80V B40950B0686M100
C1120 68uF 80V B40950B0686M100
C1121 68uF 80V B40950B0686M100
C1122 68uF 80V B40950B0686M100

BUCK12_OFF: pulled down, so a supervisor reset or hang leaves 12 V, compute and the E-stop loop running. Supervisor drives it high in Standby. BUCK24_OFF: pulled up, supervisor drives low to start 24 V.
The 12 V converter is the 300 kHz master; SYNCOUT drives the 24 V converter 180 degrees out of phase. 12 V feeds its own VCC through a Schottky after start-up (lower loss).

3.3 V always-on, on VSYS_OR after the ideal diode so it rides the hold-up bank with compute. EN tied to VIN. SNS- mode removes VSYS. 400 kHz, 3.3 V.

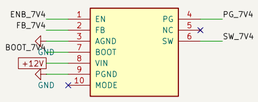
U1104
LMR38010SDDAR
L1103 XAL6060-223MEC 22uH
R1123 220k
R1124 43.2k
R1125 64.9k
C1161 4.7uF 100V
C1162 100nF 100V
C1163 100nF
C1164 22uF 25V
C1165 22uF 25V
C1166 22uF 25V
R1126 100k

LM5050-1, LM5145 x2, LMR38010. Hand design, values in RATIONALE-log.

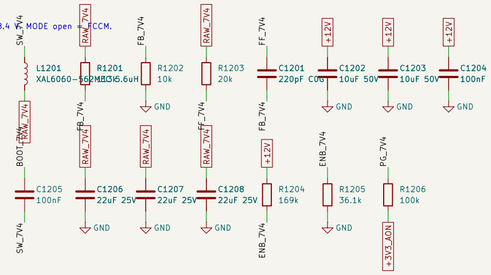
marengo		
Sheet: /LP_logic_board/front_end_12v_24v/		
File: front_end_12v_24v.kicad_sch		
Title: front_end_12v_24v: ideal diode, hold-up, 12 V, 24 V, 3.3 V always-on		
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 22/27

Servo rail. Adjust by R top: 90.9k = 6.05 V, 113k = 7.4 V, 130k = 8.4 V. MODE open = FCCM.

U1201

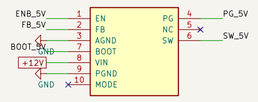


TPS56637RPAR

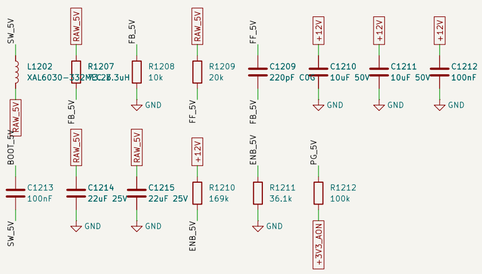


5 V peripheral, 4 A. EN divider per TI: starts near 10 V input.

U1202

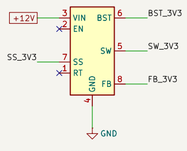


TPS56637RPAR

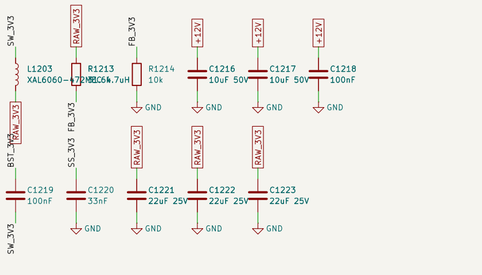


3.3 V peripheral, 2 A. RT open = 500 kHz, EN open = on.

U1203

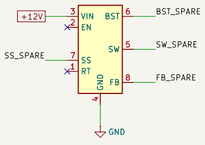


TPS62933DRLR

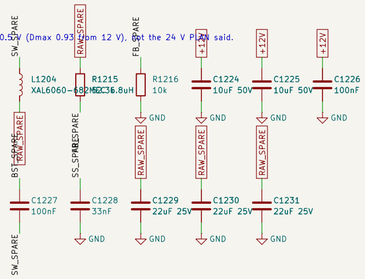


Spare rail, stuffed 5 V. 3.3 V: R top 31.6k. 9 V: see log. Max about 10.5 V (Dmax 0.93 from 12 V), but the 24 V PSU said.

U1204

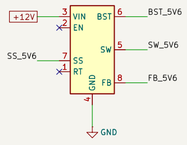


TPS62933DRLR

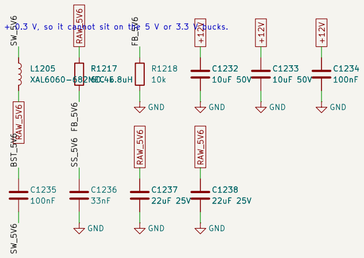


New: 5.6 V pre-regulator. TPS7A20 needs VIN <= 6.0 V and >= VOUT +0.3 V, so it cannot sit on the 5 V or 3.3 V bucks.

U1205

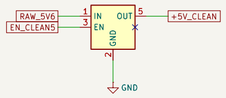


TPS62933FDRLR

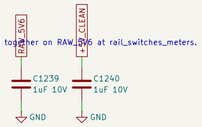


Clean rails, 0.3 A each, under 5 mV ripple. Metered together on RAW_5V6 at rail_switches_meters.

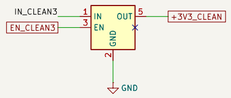
U1206



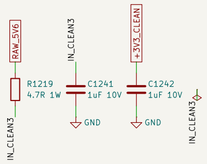
TPS7A2050PDBVR



U1207



TPS7A2033PDBVR





TPS56637 x2, TPS62933 x3, TPS7A20 x2. Hand design, values in RATIONALE-log.

marengo

Sheet: /LP_logic_board/pol_rails/

File: pol_rails.kicad_sch

Title: pol_rails: point-of-load rails from 12 V

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 23/27

OUT_24V: meter on bus A, straps A1,A0 = GND,GND, shunt 10mR

U1305
TPS26631PWPR
R1301 4.99k
R1302 475k
R1303 7.87k
R1304 22.1k
C1301 22nF
C1302 1uF 50V
C1303 100nF 50V
C1304 100nF 50V
R1305 33k
R1306 10mR 3W
R1307 10R
R1308 10R
C1305 100nF
C1306 100nF
U1306
INA238AIDGSR
OUT_24V
SP_24V
SN_24V
I2C_A_SDA
I2C_A_SCL
RAIL_ALERT_A

+12V total: converter output to rail, bus A, straps VS,VS.
ALERT at 12.0 A, latched, 150 us conversion, sheds third board, aux, spare in hardware. Firmware clears the latch.

R1368 2mR 3W
R1369 10R
R1370 10R
C1355 100nF
C1356 100nF
R1371 10k
SHED_N

OUT_12V_JETSON: meter on bus A, straps A1,A0 = GND,VS, shunt 10mR

U1303
TPS26631PWPR
R1309 6.04k
R1310 442k
R1311 21.5k
R1312 41.2k
C1307 47nF
C1308 1uF 50V
C1309 100nF 50V
C1310 100nF 50V
R1313 33k
R1314 10mR 3W
R1315 10R
R1316 10R
C1311 100nF
C1312 100nF
U1304
INA238AIDGSR
OUT_12V_JETSON
SP_12V_JETSON
SN_12V_JETSON
I2C_A_SDA
I2C_A_SCL
RAIL_ALERT_A

OUT_12V_THIRD: meter on bus A, straps A1,A0 = GND,SDA, shunt 10mR

U1305
TPS26631PWPR
R1317 6.04k
R1318 442k
R1319 21.5k
R1320 41.2k
C1313 47nF
C1314 1uF 50V
C1315 100nF 50V
C1316 100nF 50V
R1321 33k
R1322 10mR 3W
R1323 10R
R1324 10R
C1317 100nF
C1318 100nF
U1307
INA238AIDGSR
OUT_12V_THIRD
SP_12V_THIRD
SN_12V_THIRD
I2C_A_SDA
I2C_A_SCL
RAIL_ALERT_A

Q1301 2N7002
Q1302 2N7002
U1322
SN74LVC1G14DBV
SHED_12V
SHED_N

OUT_12V_AUX: meter on bus A, straps A1,A0 = GND,SCL, shunt 10mR

U1302
TPS26631PWPR
R1325 4.53k
R1326 442k
R1327 21.5k
R1328 41.2k
C1319 47nF
C1320 1uF 50V
C1321 100nF 50V
C1322 100nF 50V
R1329 33k
R1330 10mR 3W
R1331 10R
R1332 10R
C1323 100nF
C1324 100nF
U1308
INA238AIDGSR
OUT_12V_AUX
SP_12V_AUX
SN_12V_AUX
I2C_A_SCL
I2C_A_SDA
RAIL_ALERT_A

OUT_12V_PI_STUB: meter on bus A, straps A1,A0 = VS,GND, shunt 10mR

U1309
TPS26631PWPR
R1333 4.99k
R1334 442k
R1335 21.5k
R1336 41.2k
C1325 47nF
C1326 1uF 50V
C1327 100nF 50V
C1328 100nF 50V
R1337 33k
R1338 10mR 3W
R1339 10R
R1340 10R
C1329 100nF
C1330 100nF
U1310
INA238AIDGSR
OUT_12V_PI_STUB
SP_12V_PI
SN_12V_PI
I2C_A_SDA
I2C_A_SCL
RAIL_ALERT_A

OUT_7V4_SERVO: meter on bus B, straps A1,A0 = GND,GND, shunt 5mR

U1311
TPS26631PWPR
R1341 3.01k
R1342 383k
R1343 64.9k
R1344 64.9k
C1331 47nF
C1332 1uF 50V
C1333 100nF 50V
C1334 100nF 50V
R1345 33k
R1346 5mR 3W
R1347 10R
R1348 10R
C1335 100nF
C1336 100nF
U1312
INA238AIDGSR
OUT_7V4_SERVO
SP_7V4
SN_7V4
I2C_B_SDA
I2C_B_SCL
RAIL_ALERT_B

OUT_5V_PERIPH: meter on bus B, straps A1,A0 = GND,VS, shunt 5mR

U1313
TPS22965DSGR
F1301 2.5A MF-MSMF250/16X
R1350 10mR 3W
R1351 10R
R1352 10R
C1337 1uF 10V
C1338 1nF
C1339 10uF 10V
R1349 100k
C1340 100nF
C1341 100nF
U1314
INA238AIDGSR
OUT_5V_PERIPH
SP_5V
SN_5V
I2C_B_SDA
I2C_B_SCL
RAIL_ALERT_B

OUT_3V3_PERIPH: meter on bus B, straps A1,A0 = GND,SDA, shunt 10mR

U1315
TPS22965DSGR
F1302 1.1A MF-MSMF110/16X
R1354 10mR 3W
R1355 10R
R1356 10R
C1342 1uF 10V
C1343 1nF
C1344 10uF 10V
R1353 100k
C1345 100nF
C1346 100nF
U1316
INA238AIDGSR
OUT_3V3_PERIPH
SP_3V3
SN_3V3
I2C_B_SDA
I2C_B_SCL
RAIL_ALERT_B

OUT_SPARE: meter on bus B, straps A1,A0 = GND,SCL, shunt 10mR

U1317
TPS26631PWPR
R1357 4.99k
R1358 365k
R1359 33.2k
R1360 97.6k
C1347 47nF
C1348 1uF 50V
C1349 100nF 50V
C1350 100nF 50V
R1361 33k
R1362 10mR 3W
R1363 10R
R1364 10R
C1351 100nF
C1352 100nF
U1318
INA238AIDGSR
OUT_SPARE
SP_SPARE
SN_SPARE
I2C_B_SCL
I2C_B_SDA
RAIL_ALERT_B

RAW_5V6_M: meter on bus B, straps A1,A0 = VS,GND, shunt 10mR

U1319
TPS26631PWPR
R1365 10mR 3W
R1366 10R
R1367 10R
C1353 100nF
C1354 100nF
U1320
INA238AIDGSR
RAW_5V6_M
SP_CLEAN
SN_CLEAN
I2C_B_SDA
I2C_B_SCL
RAIL_ALERT_B

TPS26631 OV/OV chain mandatory (UVLO to GND means 15.46 V, would kill a 12 V rail). Latch mode, supervisor clears. SHDN default off.
TPS22965 has no current limit. Its 6 A is under the buck limit; polyfuse after each switch. 5 V 2.5 A, 3.3 V 1.1 A.
Output limits: 24 V 3.61 A, Jetson 2.98 A, third 2.98 A, aux 3.97 A, PI stub 3.61 A, 7.4 V 5.98 A, spare 3.61 A (TPS26631 latch off), 5 V 2.5 A, 3.3 V 1.2 A (polyfuse, hold at 23 C, derates hot).
INA238 has no energy registers: firmware integrates at 100 Hz, or swap to INA228 (same footprint).

R1372 100k
R1373 10k
R1374 10k

TPS26631 eFuse / TPS22965 switch + INA238 per output. 2 I2C buses, 11 meters. Hardware 12 V sh	
marengo	
Sheet: /LP_logic_board/rail_switches_meters/	
File: rail_switches_meters.kicad_sch	
Title: rail_switches_meters: rail switches and meters	
Size: A1 | Date: 2026-11-06	Rev: A-wip
KiCad E.D.A. 10.0.8	Id: 24/27

E-stop loop, 6.9 mA: +12V, 680R, NC contacts, brake thermostat, opto A, opto B, 620R, GND. No +12 V on the loop connectors: a pin-to-pin short lands on GND and opens the loop.

J2001

J2002

J2003

E-stop loop

tether, shorting dummy

wireless contact, shorting plug

U2001

ISOM8710

U2003

U2004

ISOM8710

U2006

J2004

J2005

J2006

brake thermostat NC

wireless receiver power

E-stop reset, NO button

J2009

Fans, 12 V, 25 kHz PWM open drain (polarity inverts in firmware).

J2007

J2008

Q2003

Q2004

Q2005

fan 1

2N7002

fan 2

2N7002

fan 3

2N7002

User I/O: 6-pin shell cable, the plan's 4 pins cannot carry RGB + button. Bench jumper, buzzer, charge LED, Qwiic, air NTC.

J2010

J2011

J2012

J2013

J2014

J2015

shell: button + RGB

charge LED

bench jumper

Qwiic

intake NTC

exhaust NTC

Rail outputs, one Micro-Fit each. TS2 wake: button low pulls AFE TS2 to VSS through 10k + BAT54 with the board off.

J2016

J2017

J2018

J2019

J2020

J2021

J2022

J2023

J2024

J2025

J2026

OUT_24V

OUT_12V_JETSON

OUT_12V_THIRD

OUT_12V_AUX

OUT_12V_PI_STUB

OUT_7V4_SERVO

OUT_5V_PERIPH

OUT_3V3_PERIPH

OUT_SPARE

5V_CLEAN

3V3_CLEAN

R2001 680R 1W

R2002 620R 0.25W

D2001 1.5KE48W

D2002 SMAJ15A

D2003 SMAJ15A

C2001 100nF

C2002 100nF

C2003 100nF

C2004 100nF

F2001 0.5A polyfuse

SN74LVC1G14DBVR

SN74LVC1G74DCUR

SN74LVC1G08DBVR

2N7002

D2007 1.5KE48W

Q2006 2N7002

LS2001 PS1240P02BT

U2031 TPS3431SDRBR

U2032 SN74LVC244APWR

SUP_OPEN high: 2N7002 sinks EN. MCU dead or Hi-Z: pull-up opens the banks. The FET only sinks, so firmware cannot close an open loop.
Cat 1 option: remove the 10k filter, fit the four DNP parts, about 302 ms with the 100 nF left in (242-372 ms). DNP in rev A: Cat 0.
Re-close latch: loop open, watchdog or power-up clears the flops. Re-arming needs a rising edge from the NO reset button with the loop closed. A held button does nothing. No firmware path.
10k + 100 nF ahead of the Schmitt input rides through contact chatter, 0.6 to 0.9 ms added to a stop.
CLR is OUT AND WD_N through a gate, a clean edge: TPS3431 clears the latch at power-up and on timeout, and no capacitor sits on WD_N.
One series loop still feeds both channels: a short across one contact pair defeats both. Not Cat 3.

Hardware watchdog and port gate. Starts itself at power-up, no firmware can start or stop it:
SET1 high, CWD open = 1.6 s (1.36-1.84 s). Timeout pulls WD_N low, clears both E-stop latches.
PORT_OE_N follows latch A, so every PORT_EN tri-states and its 100k pulls it off. Compute rails untouched.

E-stop path: loop current to opto to inverter to gate-driver enable. No firmware in that path.

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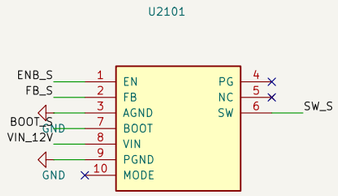
Sheet: /LP_logic_board/io_fans_estop_loop/
File: io_fans_estop_loop.kicad_sch

Title: io_fans_estop_loop: E-stop loop, fans, user I/O, output connectors

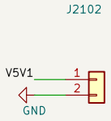
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 25/27



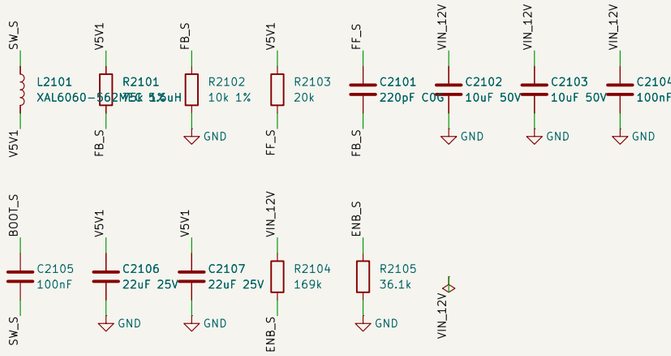
12 V in



TPS56637RPAR



5.1 V out, 8.5 A per contact



Output pigtail goes to a USB-C plug. A 2x3 housing on the GPIO header would put all 6 A through one GND pin (pin 6) and cannot lock.
Set usb_max_current_enable=1 on the Pi. PG left open: no logic rail on the stub.

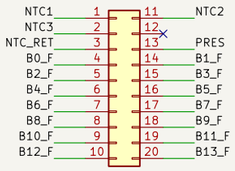
Own 2-layer board. Local nets; harness map checked in check.py.

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Sheet: /pi_power_stub/
File: pi_power_stub.kicad_sch

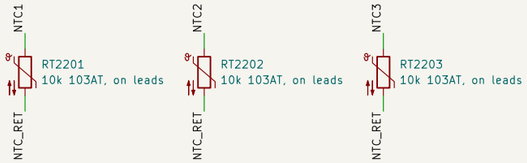
Title: pi_power_stub: 12 V to 5.1 V 6 A at the Pi

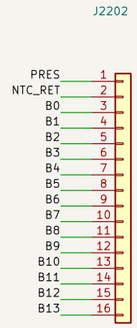
Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 26/27











Fuse at the cell, one per tap. Schurter USE 2410 1 A: 79 mOhm, 200 A at 86 VDC, tau under 0.05 ms, against 146 A prospective. Balance adds 24 mV of sense error at 150 mA across two taps.
Footprint is the Schurter land: 2.5 x 2.8 mm pads, 3.0 mm gap (data sheet p.2).
Presence: pin 13 goes out through the cell tap plug (J2202 pins 1, 2 looped in the harness, beside B0, away from B13) and back on pin 3, so either plug out opens DSG and CHG. B0 and B13 are their own Kelvin tabs: no pack current in a sense wire. NTC_RET returns on pin 3 to VSS, no cell connection.

Own 2-layer board at the pack. Pin map must match HP J2. Local nets; harness map checked in che

marengo

Sheet: /pack_sense_board/
File: pack_sense_board.kicad_sch

Title: pack_sense_board: tap fuses, NTC, presence loop

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 27/27